# EVisionAI: ML-Based EV Charging Station Recommendation and Real-Time Slot Forecasting

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***Abstract* : Electric vehicle (EV) navigation systems typically provide route guidance and station locations, but they seldom optimize charging decisions under jointly varying traffic, station congestion, battery constraints, and charger compatibility. This paper presents EVisionAI, a research-oriented EV charging decision support framework that integrates battery-aware routing, station discovery, and machine-learning-based wait-time forecasting into a unified recommendation pipeline. The backend is implemented as RESTful services (FastAPI) with WebSocket streams for live updates, while the predictive layer uses XGBoost with structured feature engineering over temporal, station-capacity, traffic, and contextual variables. To address cold-start deployment, we construct a synthetic-but-structured dataset (15,000 samples) using realistic temporal demand profiles, port-capacity constraints, and stochastic queue effects. Comparative experiments against Random Forest, Linear Regression, and SVR show that XGBoost achieves the best regression performance ($R^2$ = 0.847, MAE = 3.2 min, RMSE = 4.8 min). We further define an adaptive multi-criteria station scoring function that combines predicted wait time, detour distance, and station reliability to produce ranked, actionable recommendations. Results indicate that EVisionAI provides a reproducible and extensible framework for real-time EV charging support, while also identifying external-validity limitations and pathways for real-world calibration.**

**Keywords: electric vehicle charging intelligence, wait-time forecasting, battery-aware routing, multi-criteria decision scoring, XGBoost regression, real-time recommendation systems.**

## *I.INTRODUCTION*

Electric vehicle (EV) adoption is rapidly increasing worldwide, driven by sustainability goals and advancements in battery technology [8], [10]. However, practical challenges still limit seamless EV usage, particularly in long-distance travel. Drivers frequently face difficulties such as locating compatible charging stations, estimating waiting times, planning routes under battery constraints, and dealing with dynamic traffic conditions [9], [10]. These challenges contribute to range anxiety and inefficient trip planning.

Existing navigation systems primarily optimize routes based on distance or time but fail to incorporate EV-specific constraints such as battery state-of-charge, charger compatibility, station congestion, and charging duration [8], [9]. Similarly, most EV charging applications provide static station information with limited support for real-time decision-making or predictive insights. Recent research has explored queueing-theoretic models,

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machine learning–based forecasting, and optimization techniques for EV charging infrastructure [1], [2], [3]. However, significant gaps remain in integrating these approaches into a unified, real-time, user-centric system that combines routing, station intelligence, and predictive analytics.

To address these limitations, this paper proposes EVisionAI, an intelligent EV charging management platform that integrates route planning, charging station discovery, and wait-time prediction into a single pipeline. The system leverages machine learning models, real-time data sources, and user preferences to generate actionable charging recommendations. Unlike traditional systems, it incorporates predictive wait-time estimation and multi-factor decision scoring to enhance usability and efficiency.

The proposed system is designed with a modular and extensible architecture, enabling seamless integration of external APIs, real-time updates, and adaptive learning components. It supports battery-aware routing, dynamic station ranking, and interactive visualization, ensuring that users receive optimized and context-aware recommendations.

This work presents a unified EV pipeline that integrates route planning, charging station discovery, and wait-time prediction into a single intelligent framework. It introduces an adaptive scoring mechanism for ranking charging stations based on wait time, distance, reliability, and battery constraints. Additionally, a comparative machine learning evaluation demonstrates the effectiveness of XGBoost over Random Forest, Linear Regression, and Support Vector Regression for wait-time prediction. A structured synthetic data generation strategy is also employed to enable model training and deployment in the absence of real-world datasets.

## *II. LITERATURE REVIEW*

### A. Queueing-Theoretic and Stochastic Modeling

Queueing theory has been widely applied to model EV charging station behavior, particularly in analyzing congestion, service rates, and user dynamics. Prior studies utilize stochastic approaches such as continuous-time Markov chains to represent multi-server charging environments while incorporating realistic user behaviors like reneging and balking [1]. These models provide strong analytical capabilities for deriving performance metrics such as system occupancy, server utilization, and waiting probabilities.

However, such approaches often rely on simplifying assumptions and are computationally expensive when applied to large-scale real-world systems. Additionally, they are primarily suited for offline analysis and lack integration with real-time decision-making frameworks. Hybrid approaches combining queueing theory with deep learning improve estimation efficiency [3], but they remain focused on infrastructure planning rather than real-time user-centric applications.

EVisionAI addresses these limitations by leveraging queueing-theoretic insights within a real-time predictive framework, enabling scalable and practical wait-time estimation for end users.

### B. Machine Learning for Load and Wait-Time Forecasting

Machine learning techniques have been increasingly used for EV charging demand and wait-time prediction. Advanced ensemble methods, such as hybrid architectures combining LSTM networks with gradient boosting models, have demonstrated improved forecasting accuracy for charging load and grid demand [2]. Similarly, Artificial Neural Networks (ANNs) have been applied for real-time optimization of charging operations under simulated conditions [5].

Despite their strong predictive capabilities, these approaches are often limited to grid-level forecasting or station-level optimization, rather than user-centric recommendation systems. Many models also depend heavily on large-scale real-world datasets, making them less effective in cold-start scenarios. Furthermore, limited comparative evaluation across models reduces clarity in selecting the most suitable approach.

EVisionAI overcomes these challenges by conducting a comparative evaluation of multiple regression models and demonstrating the effectiveness of XGBoost for wait-time prediction. Additionally, a structured synthetic-data generation protocol is introduced to ensure reliable performance in data-scarce environments.

### C. Routing and Recommendation Systems

Routing and recommendation systems for EVs have evolved from basic navigation tools to more intelligent platforms incorporating charging station discovery and optimization. Existing systems provide functionalities such as nearest station identification and real-time slot booking [9]. Optimization-based methods, including multi-objective techniques, have also been explored for infrastructure planning and grid performance improvement [6].

However, most existing solutions treat routing, charging, and recommendation as separate problems. They lack integration of real-time constraints such as battery state, waiting time, and traffic conditions into a unified decision-making framework. Additionally, many systems rely on static rules or single-factor optimization, limiting their adaptability in dynamic environments.

EVisionAI introduces a unified pipeline that integrates routing, station discovery, and wait-time prediction into a single framework. By incorporating battery-aware routing and real-time data, the system provides adaptive and context-aware recommendations.

### D.Research Gap and EVisionAI Positioning

A review of existing literature reveals several key gaps. Most studies focus on isolated components such as queue modeling, load forecasting, or infrastructure planning without integrating them into a unified real-time system. There is limited use of multi-criteria decision-making approaches that simultaneously consider distance, waiting time, and reliability. Additionally, the cold-start problem remains a challenge due to limited availability of real-world datasets, and few works provide comparative evaluation of machine learning models for EV charging recommendations.

EVisionAI addresses these gaps by proposing an end-to-end intelligent EV charging recommendation system that integrates routing, real-time station discovery, and machine learning-based wait-time prediction. The system employs an adaptive multi-

criteria scoring mechanism for decision-making, validates performance through comparative model analysis, and incorporates a structured synthetic-data protocol to ensure robustness in data-scarce scenarios.

## *III. SYSTEM ARCHITECTURE AND DESIGN*

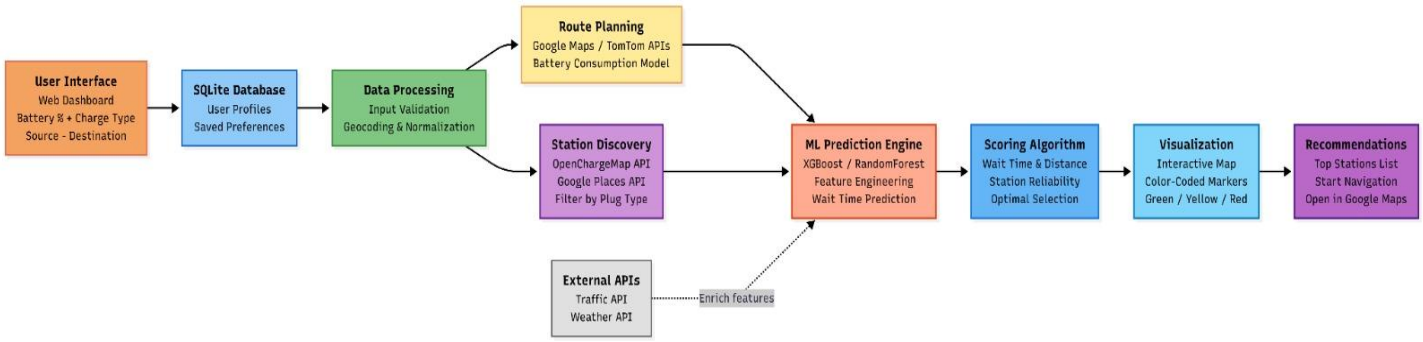


Fig.1.Proposed system architecture of the EVisionAI intelligent EV charging management platform

Fig.1.describes the proposed system architecture of the EVisionAI intelligent EV charging management platform. This architecture describes an end-to-end EV charging recommendation system that turns user inputs and external data into intelligent, map-based suggestions for where to charge. The pipeline starts with the User Interface, typically a web dashboard or mobile app. Here the driver enters their battery percentage, charge type (fast/slow or plug type), source, and destination. This UI is responsible only for collecting clean, structured inputs and displaying results; all heavy logic is delegated to the backend. Next, the data goes into the SQLite Database. This layer stores user profiles and saved preferences, such as preferred charger type, home location, or frequently used routes. Persisting this information allows the system to auto-fill forms, personalize recommendations (for example, bias towards a user's favorite network), and support analytics over time. SQLite is a lightweight, file-based database that fits well for a single-node backend or prototype. The backend uses RESTful JSON APIs over HTTP(S), implemented using FastAPI, for handling station discovery, wait-time estimation, route planning, and traffic prediction. Additionally, WebSocket communication is used to enable real-time updates and live data streaming.

From there, the input flows into Data Processing. This block performs input validation (checking that battery percentage is 0–100, addresses are non-empty, charge type is recognized, etc.) and geocoding & normalization. Geocoding converts textual locations into latitude/longitude using services like Google or OpenStreetMap; normalization ensures that units (km, minutes, %, kWh) and formats are consistent. This step is critical to avoid propagating bad data into downstream ML models or APIs. After preprocessing, the pipeline branches into two main analytical paths. The first is Route Planning, which uses Google Maps / TomTom APIs to compute driving routes and distances between the source and destination. On top of the raw route, a Battery Consumption Model estimates how much battery will be used along the trip, considering distance and typical consumption (e.g., kWh per 100 km). This makes it possible to decide whether intermediate charging stops are needed, and where along the route they should ideally occur.

The second analytical branch is Station Discovery. Using OpenChargeMap and optionally Google Places, the system retrieves nearby charging stations around the route or the user's current position. It then filters by plug type (e.g., CCS2, CHAdeMO, Type-2) so that only compatible stations are kept. At this stage, stations are enriched with metadata such as number of ports, operator, and approximate location accuracy. Both the route and station information feed into the ML Prediction Engine. This component encapsulates model built with XGBoost or RandomForest and associated feature engineering. Features typically include time of day, day of week, station type, number of ports, historical usage, distance from the driver, and possibly traffic or weather signals. The main output is a predicted wait time at each station—how long a driver is likely to wait before getting a free charger—plus an uncertainty or confidence score.

Predicted wait times are then passed into the Scoring Algorithm. This block combines wait time, distance, and station reliability (e.g., based on historical uptime, number of ports, or user feedback) into a single optimality score for each station. The scoring function penalizes long detours and long waits, and rewards reliable, high-capacity stations close to the planned route, optionally adjusting weighting when battery is very low. The ranked results move into the Visualization component. Here, an interactive map displays stations as color-coded markers—typically green for low wait time.Tooltips can show exact wait time, distance, and plug types. This visual layer helps users quickly grasp their options along the route.

Finally, the Recommendations block presents a top stations list alongside the map. For each recommended station, the system provides "Start Navigation" and "Open in Google Maps" links, so the user can hand off the chosen station and route to their

preferred navigation app. Together, these stages form a coherent pipeline: from user input and stored preferences, through validation, route and station analytics, machine-learning-driven wait-time prediction, multi-criteria scoring, and ending in a clear, actionable set of EV charging recommendations.

Predicted wait times are then passed into the Scoring Algorithm, where stations are ranked using a multi-criteria function:

$$Score(s) = w_t T_{wait}(s) + w_d D_{detour}(s) + w_r (1 - R_{rel}(s)) + w_c C_{compat}(s) \quad (1)$$

The station ranking is computed using Equation (1).

where $T_{wait}$ is predicted wait time, $D_{detour}$ is detour cost, $R_{rel}$ is reliability, and $C_{compat}$ is compatibility penalty. Lower score indicates a better station.

When battery is low (e.g., <20%), higher weights are assigned to wait time and detour distance to prioritize nearby stations with shorter queues .

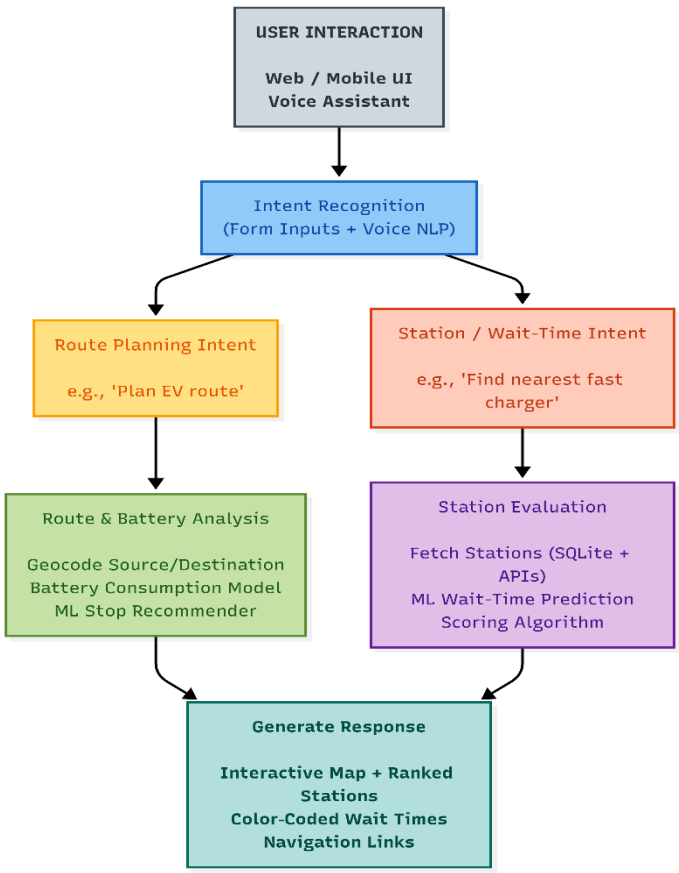


Fig.2.System Workflow System

Fig. 2 describes the overall workflow of the proposed EV charging system, starting from user interaction and intent recognition to route analysis or station evaluation, and finally generating an interactive map with ranked stations and wait-time predictions.

Workflow Architecture Overview

The workflow processes user requests through intent recognition, routes them to specialized modules, and returns actionable recommendations.

Components and Flow

1. User Interaction

Entry point: Web/Mobile UI and Voice Assistant. Users submit route planning queries, station searches, or wait-time checks via forms or voice.

2. Intent Recognition

Analyzes input to determine intent:

Form inputs: structured fields (battery %, charge type, source, destination)

Voice NLP: natural language understanding for voice commands

Outputs: Route Planning Intent or Station/Wait-Time Intent.

3. Route Planning Intent Branch

Triggered by queries like "Plan EV route" or "Find route from A to B."

Route & Battery Analysis module:

Geocoding: converts addresses to coordinates

Battery Consumption Model: estimates energy use based on distance, elevation, speed, and vehicle specs

ML Stop Recommender: suggests optimal charging stops using XGBoost/RandomForest, considering battery level, station availability, and route efficiency

4. Station/Wait-Time Intent Branch

Triggered by queries like "Find nearest fast charger" or "Show stations with low wait time."

Station Evaluation module:

Fetch Stations: combines SQLite (user preferences, historical data) with external APIs (OpenChargeMap, Google Places) and filters by plug type

ML Wait-Time Prediction: predicts wait times using features like time of day, day of week, station capacity, historical patterns, traffic, and weather

Scoring Algorithm: ranks stations by wait time, distance, reliability, and user preferences

5. Generate ResponseConverges outputs from both branches into a unified response:

Interactive Map: visual display with route (if applicable) and charging stations

Ranked Stations List: top recommendations with details (name, distance, wait time, charger types)

Color-Coded Wait Times: green (<5 min), yellow (5–15 min), red (>15 min)

Navigation Links: direct links to Google Maps or other navigation apps

This workflow transforms raw user input into personalized, data-driven EV charging recommendations through intent recognition, specialized processing modules, and unified response generation

6.Voice NLP Processing

The voice interaction module follows a structured Natural Language Processing (NLP) pipeline. First, user speech is converted into text using a speech-to-text engine. The transcribed text is then processed using intent classification to identify the user's request, such as RoutePlanning, StationSearch, or WaitTimeQuery.

Next, slot extraction is performed to identify key parameters including source, destination, charger type, and battery level. A

confidence score is generated for each prediction. If the confidence falls below a predefined threshold, the system falls back to manual form input to ensure reliability.
This approach is chosen for its low latency, deterministic mapping to backend APIs, and improved robustness, enabling accurate and real-time voice-driven interactions.

## *IV. System Implementation*

EVisionAI follows a modular architecture separating data processing, machine learning inference, and user interface components. The backend is implemented using FastAPI (Python) for high performance, asynchronous request handling, and automatic API documentation. The waiting time prediction model is trained using the XGBoost library. The feature engineering pipeline applies transformation functions to convert raw inputs into a structured feature matrix. Categorical variables are encoded using pandas.get_dummies(), while numerical features are normalized using StandardScaler, fitted on training data and applied to test data to prevent data leakage.The XGBoost model uses the following hyperparameters: n_estimators = 100, max_depth = 6, learning_rate = 0.1, and random_state = 42. Training is performed on 12,000 samples (80%), with evaluation on 3,000 samples (20%). Early stopping is applied, halting training if validation loss does not improve for 10 consecutive epochs. During inference, real-time requests are processed by extracting contextual features, applying the same preprocessing steps used in training, and passing the feature vector to the trained model. The predicted waiting time (in minutes) is categorized into green (<5 minutes), yellow (5–15 minutes), or red (>15 minutes) for intuitive visualization.

Synthetic Data Generation for Cold-Start
To address the lack of real-world datasets during initial deployment, a structured synthetic data generation approach is used for training the machine learning models.
Temporal patterns are simulated using realistic distributions, incorporating hourly and daily demand variations (e.g., peak usage during morning and evening hours, lower demand at night). Station-specific features such as number of charging ports, charger type (fast/slow), and reliability scores are generated based on practical ranges observed in real EV infrastructure.
The demand process is modeled using probabilistic assumptions, where vehicle arrivals follow time-dependent distributions to reflect real-world usage behavior. Queue-capacity constraints are enforced by limiting the number of simultaneous charging sessions based on available ports, ensuring realistic waiting time generation.
To improve model generalization, controlled noise is injected into features such as arrival rates, service times, and station availability. Additionally, all generated values are clipped within realistic bounds (e.g., wait time ≥ 0, battery percentage between 0–100) to maintain data validity.
This structured synthetic data generation process enables robust model training under cold-start conditions and supports reliable performance metrics, including the reported $R^2$ score.

Evaluation Protocol
To ensure robust model evaluation, a 5-fold cross-validation strategy is applied on the training dataset instead of relying solely on a single 80/20 split. This improves generalization and reduces overfitting bias.
Hyperparameter tuning is performed using grid search to identify the optimal model configuration. The parameter grid used for XGBoost is as follows:
n_estimators = {100, 200, 300}
max_depth = {4, 6, 8}
learning_rate = {0.03, 0.1, 0.2}
subsample = {0.8, 1.0}
colsample_bytree = {0.8, 1.0}

The final hyperparameters are selected based on the best average validation performance across all folds. The optimized model is then evaluated on a held-out test set to report the final performance metrics.This evaluation protocol ensures reliable performance estimation and improves the validity of the reported results.

Performance Metrics
Model performance is evaluated using standard regression metrics:
MeanAbsoluteError(MAE)
*(1) Mean Absolute Error:*

$$MAE = \left(\frac{1}{n}\right) * \sum_{i=1}^{n} |y_i - \hat{y}i|$$

where n is the number of samples, $y_i$ is the actual waiting time, and $\hat{y}_i$ is the predicted waiting time.

Mean Squared Error (MSE)
*(2) Mean Squared Error:*

$$MSE = \left(\frac{1}{n}\right) * \sum_{i=1}^{n} (y_i - \hat{y}i)^2$$

Root Mean Squared Error
*(3) Root Mean Squared Error:*

$$RMSE = \sqrt{\{MSE\}}$$

$$RMSE = \sqrt{\left(\frac{1}{n}\right) * \sum_{i=1}^{n} (y_i - \hat{y}i)^2}$$

Coeffcient of Determination ($R^2$ Score)
*(4) $R^2$ Score:*

$$R^2 = 1 - \left(\frac{SS_{res}}{SS_{tot}}\right)$$

where:

$$SS_{res} = \sum (y - \hat{y}i)^2$$

$$SS_{tot} = \sum(y_i - \bar{y})^2$$

and $\bar{y}$ is the mean of actual waiting times.
The model achieves $R^2 = 0.847$, MAE = 3.2 minutes, and RMSE = 4.8 minutes, indicating strong predictive performance and reliable waiting time estimation.

Table .1. Comparative analysis of ML techniques with different performance metrics.

| Performance Metric | XGBoost Regressor | Random Forest Regressor | Linear Regression | Support Vector Regression |
|---|---|---|---|---|
| MAE (minutes) | 3.2 | 4.1 | 6.8 | 5.3 |
| MSE | 23.04 | 28.81 | 78.4 | 42.25 |
| RMSE (minutes) | 4.8 | 5.4 | 8.9 | 6.5 |
| $R^2$ Score | 0.847 | 0.809 | 0.623 | 0.742 |
| Training Time (seconds) | 12.5 | 8.3 | 0.5 | 15.2 |
| Prediction Latency (ms) | 8.2 | 12.1 | 2.1 | 18.5 |

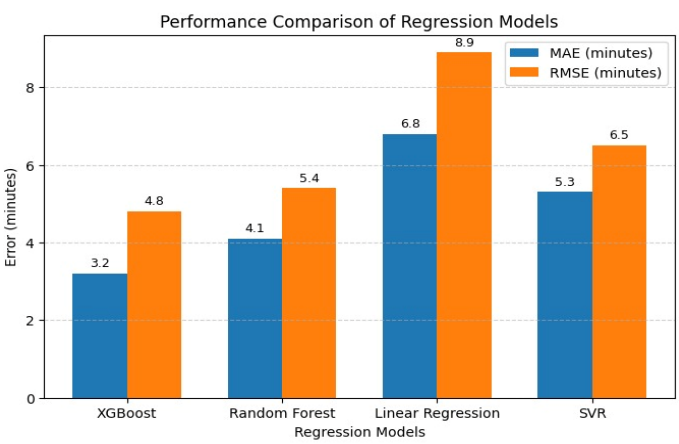


Fig.3.Performance Comparison of Regression Models Based on MAE and RMSE

Table .1 & Fig. 3 describes the comparative performance of different regression models using multiple evaluation metrics. The comparative results indicate that the XGBoost Regressor achieves the best overall performance, with the lowest MAE (3.2 minutes), lowest RMSE (4.8 minutes), and highest $R^2$ score (0.847), demonstrating superior prediction accuracy. Random Forest performs competitively but slightly below XGBoost. Linear Regression shows the weakest predictive performance despite minimal training time. Support Vector Regression provides moderate accuracy but incurs higher training and prediction latency. Overall, XGBoost offers the best balance between accuracy and computational efficiency for waiting time estimation.

**Comparison of Features Between Traditional EV Charging Systems and EVisionAI**

Table .2. Comparison of Traditional Systems & EVisionAI

| **Feature** | **Traditional Systems** | **EVisionAI** |
|---|---|---|
| Charging station discovery | Yes | Yes |
| Battery-aware route planning | No | Yes |
| Wait-time prediction | No | Yes |
| ML-based decision making | No | Yes |
| Traffic delay integration | No | Yes |
| Voice-assisted interaction | No | Yes |
| Real-time updates | Limited | Yes |

Table. 2 describes the feature-wise comparison between traditional EV charging systems and EVisionAI.The comparison highlights that while traditional EV charging systems support basic charging station discovery, they lack advanced intelligent features. EVisionAI enhances functionality by integrating battery-aware route planning, wait-time prediction, machine learning–based decision making, traffic delay integration, voice-assisted interaction, and real-time updates. Overall, EVisionAI provides a more intelligent, adaptive, and user-centric charging management solution compared to conventional systems.
Comparison of ML Models

Table 3. Comparison of Machine Learning Models

Before evaluating classification performance, the predicted wait times are categorized into discrete classes for decision-making. The classification labels are defined as follows:

Green: wait time < 5 minutes
Yellow: wait time between 5–15 minutes
Red: wait time > 15 minutes

The classification metrics such as Accuracy, Precision, Recall, and F1-Score reported in this table correspond to the above wait-time categorization task.

| **Metric** | **XGB (Wait)** | **RF (Wait)** | **XGB (Traffic)** | **RF (Traffic)** | **XGB (Route)** | **RF (Route)** | **Heuristic** |
|---|---|---|---|---|---|---|---|
| **Accuracy** | **89.25** | **87.50** | **91.20** | **88.75** | **92.15** | **89.30** | **76.40** |
| **Precision** | **88.00** | **87.00** | **90.00** | **88.00** | **91.00** | **89.00** | **75.00** |
| **Recall** | **88.50** | **87.00** | **90.50** | **88.50** | **92.00** | **89.00** | **76.00** |

| Metric | XGB (Wait) | RF (Wait) | XGB (Traffic) | RF (Traffic) | XGB (Route) | RF (Route) | Heuristic |
|---|---|---|---|---|---|---|---|
| **F1-Score** | **88.25** | **87.00** | **90.25** | **88.25** | **91.50** | **89.00** | **75.50** |

Table 3 presents the performance comparison of machine learning models across multiple evaluation metrics. The results show that XGBoost consistently outperforms Random Forest and heuristic approaches in terms of accuracy, precision, recall, and F1-score across wait-time, traffic, and route prediction tasks. This indicates that XGBoost is more effective in capturing complex patterns and provides more reliable predictions for EV charging recommendations.

## *V.Conclusion*

This paper presented EVisionAI, an intelligent EV charging recommendation system that integrates route planning, real-time station discovery, and machine learning-based wait-time prediction. The proposed system demonstrates effective performance in predicting wait times and generating optimal charging recommendations, highlighting the advantages of using advanced regression models such as XGBoost within a unified decision framework.

The study has certain limitations. The use of synthetic data for training and evaluation may not fully capture real-world variability, which can affect external validity. Additionally, the system has not yet been validated on large-scale real-world deployments, and certain factors such as cross-network integration and dynamic pricing have not been considered.

Future work will focus on extending the system with real-world data integration, improving personalization based on user behavior, and incorporating additional factors such as pricing and charging network interoperability to enhance recommendation quality and practical applicability.

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